

**From Person-Specific Dynamics to Population Inference: A Meta-Analytic
Structural Equation Modeling Framework for Multilevel Dynamic Models**

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Computations for this research were performed on the Pennsylvania State University's Institute for Computational and Data Sciences' Roar supercomputer using SLURM for job scheduling (Yoo et al., 2003), GNU Parallel to run the simulations in parallel (Tange, 2021), and Apptainer to ensure a reproducible software stack (Kurtzer et al., 2017, 2021).

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Links

Research Compendium

The data and materials for this study are available on OSF (<https://osf.io/uenr7>) and GitHub (<https://github.com/jeksterslab/manMetaVAR>, <https://jeksterslab.github.io/manMetaVAR/index.html>).

fitVARMxID R Package

Source code and documentation for the `fitVARMxID` R package are available on GitHub (<https://github.com/jeksterslab/fitVARMxID>, <https://jeksterslab.github.io/fitVARMxID/index.html>).

metaDyn R Package

Source code and documentation for the `metaDyn` R package are available on GitHub (<https://github.com/jeksterslab/metaDyn>, <https://jeksterslab.github.io/metaDyn/index.html>).

Single Replication from the Simulation Study

<https://jeksterslab.github.io/manMetaVAR/articles/sim-rep.html>

Empirical Illustration: Meta-Analysis of Affect Dynamics

<https://jeksterslab.github.io/manMetaVAR/articles/empirical.html>

Benchmarking MetaVAR Against Naive Two-Stage Estimation and Bayesian Multilevel VAR

<https://jeksterslab.github.io/manMetaVAR/articles/benchmark.html>

Containers for Reproducibility

<https://jeksterslab.github.io/manMetaVAR/articles/containers.html>

Figure 1
Relative Bias for the Fixed Effects

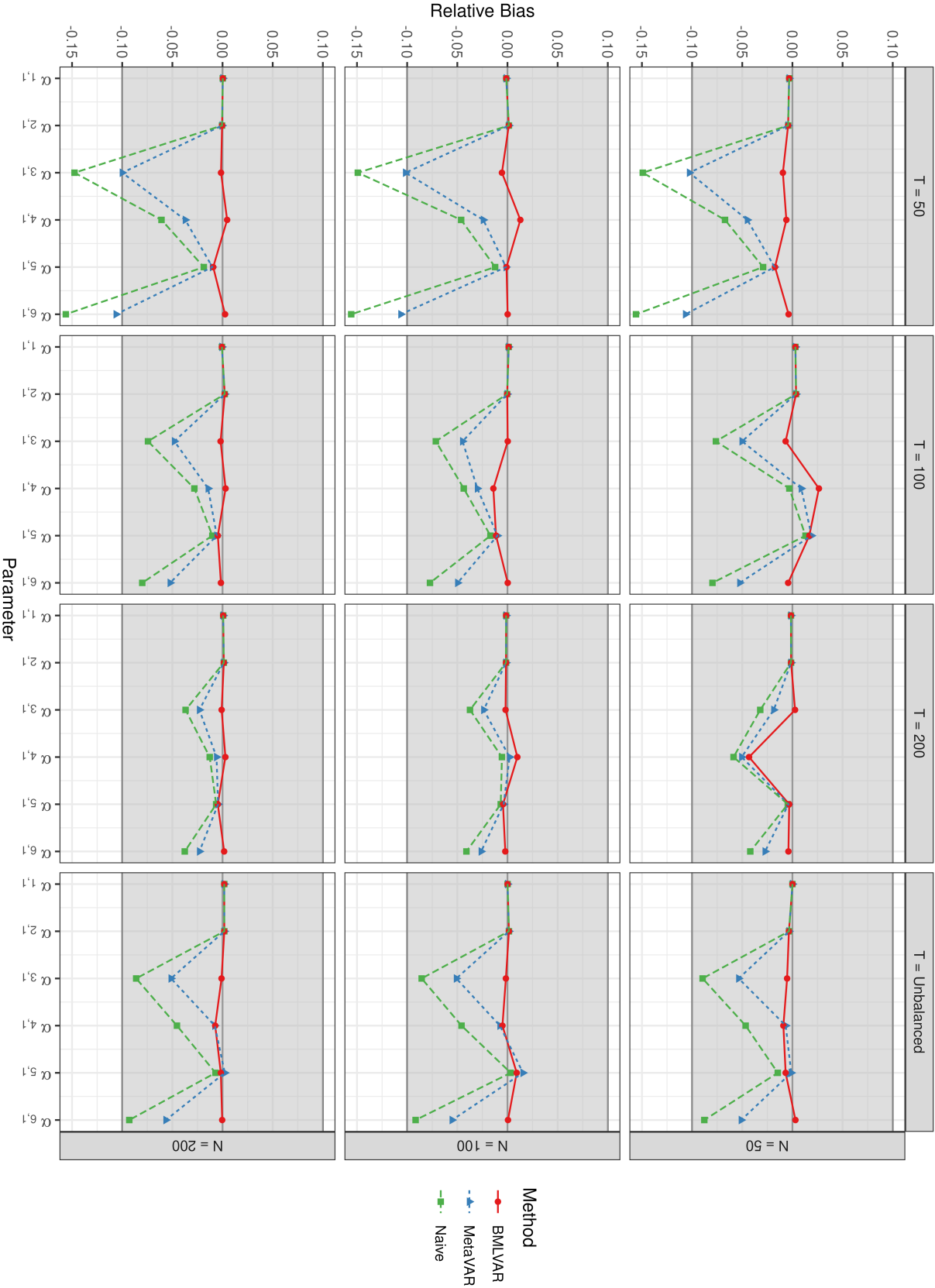


Figure 2
Relative Bias for the Random Effects

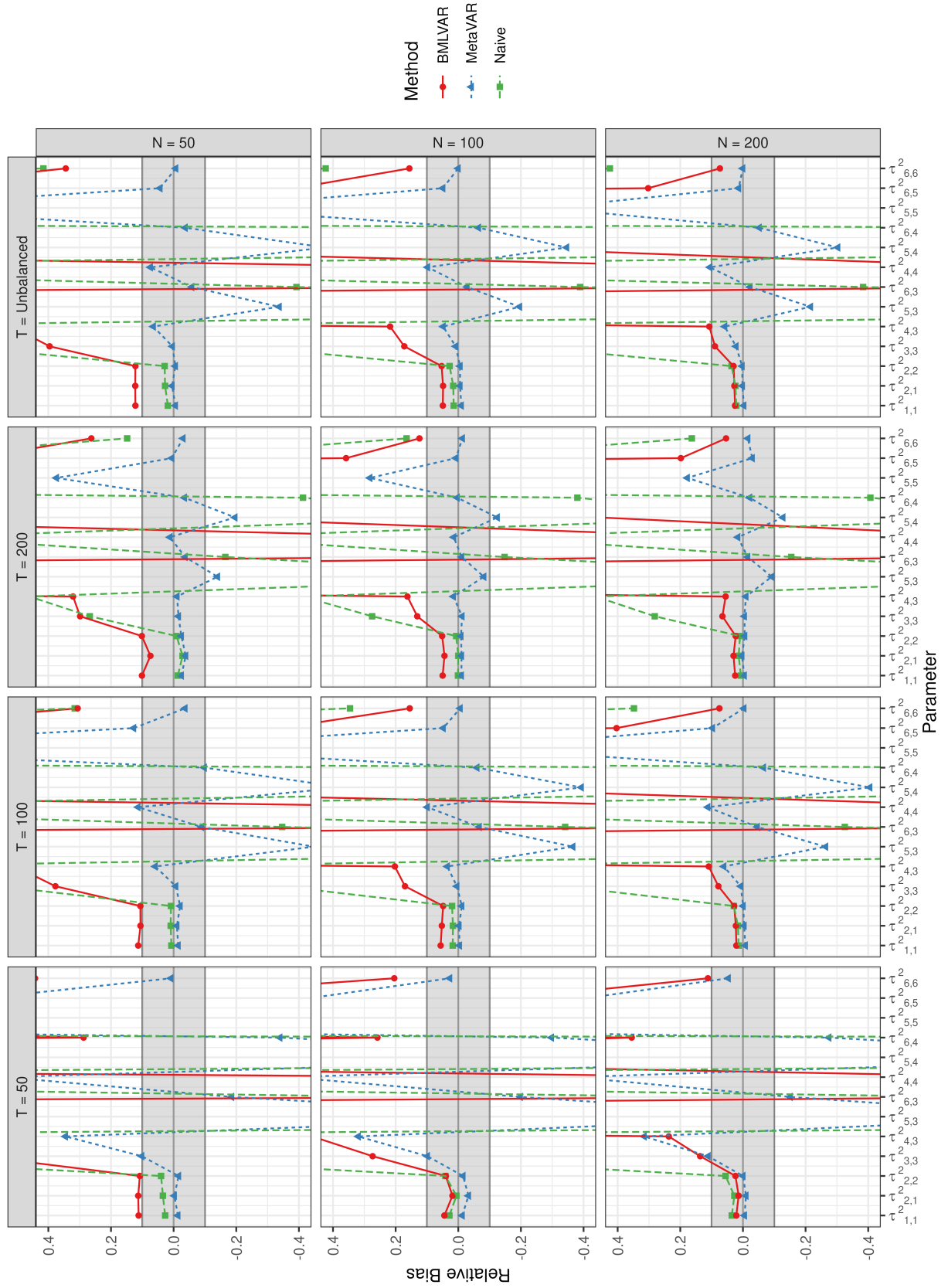
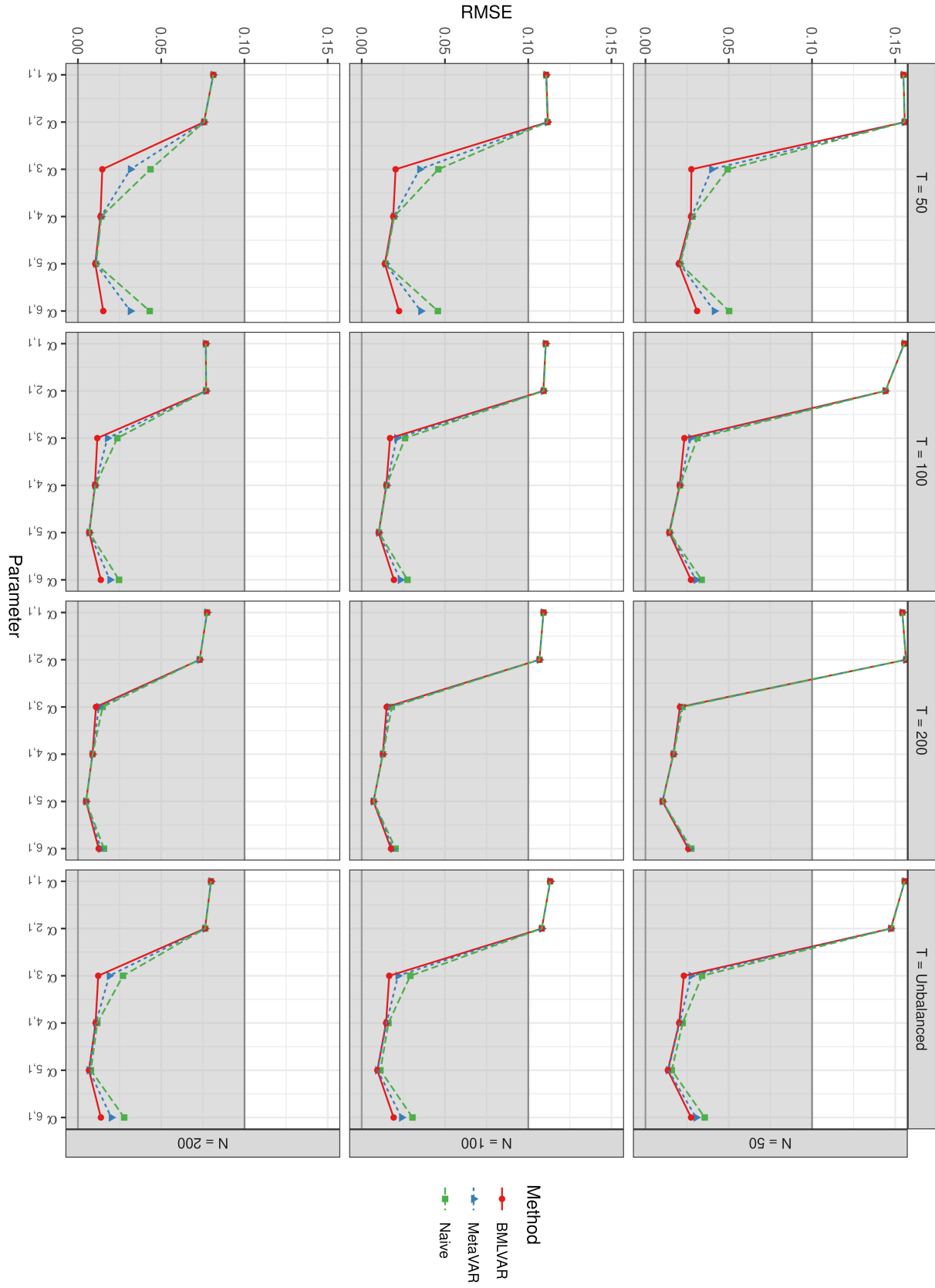


Figure 3
Root Mean Square Error for the Fixed Effects



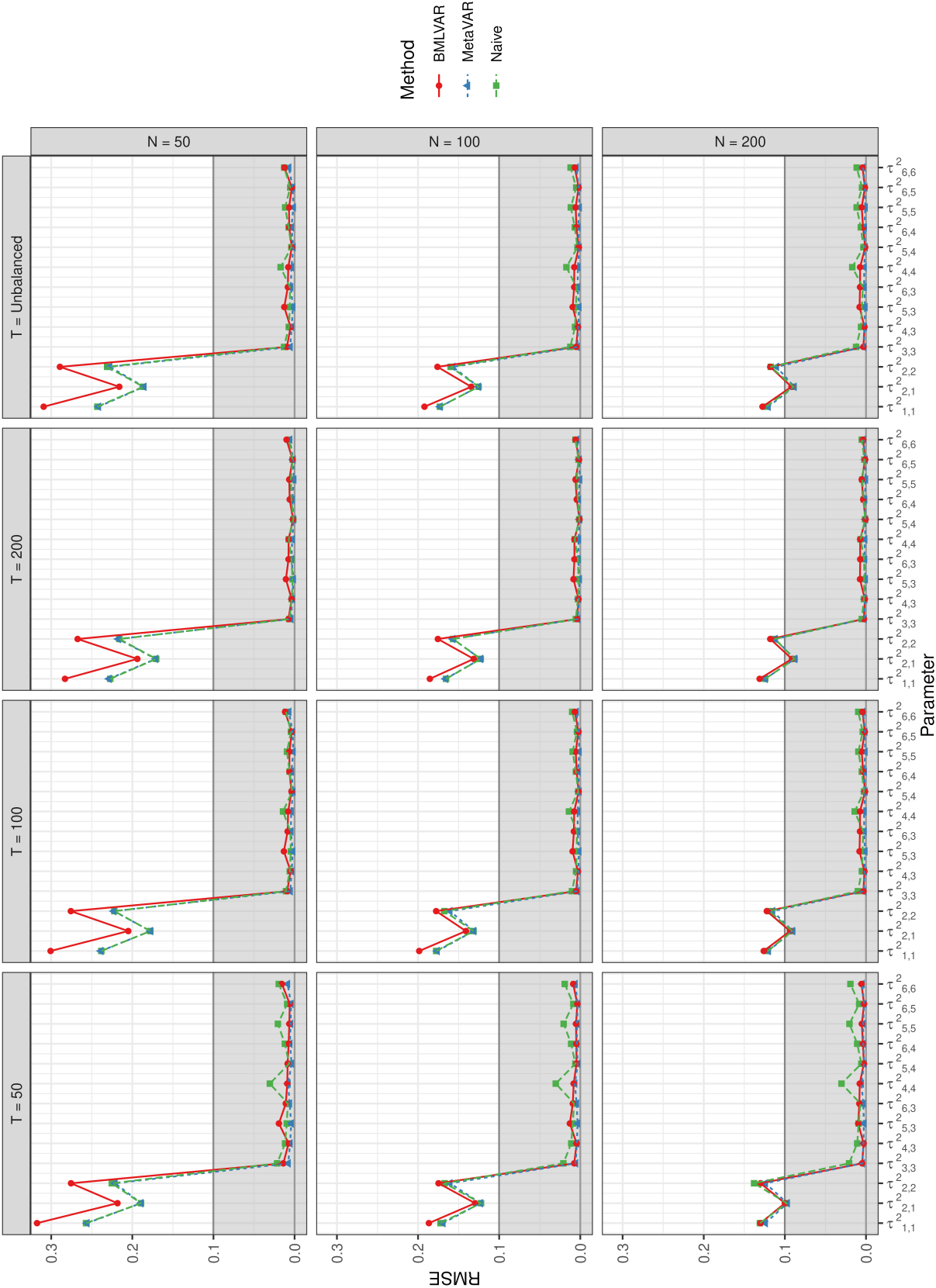


Figure 4
Root Mean Square Error for the Random Effects

Figure 5
Coverage Probability for the Fixed Effects

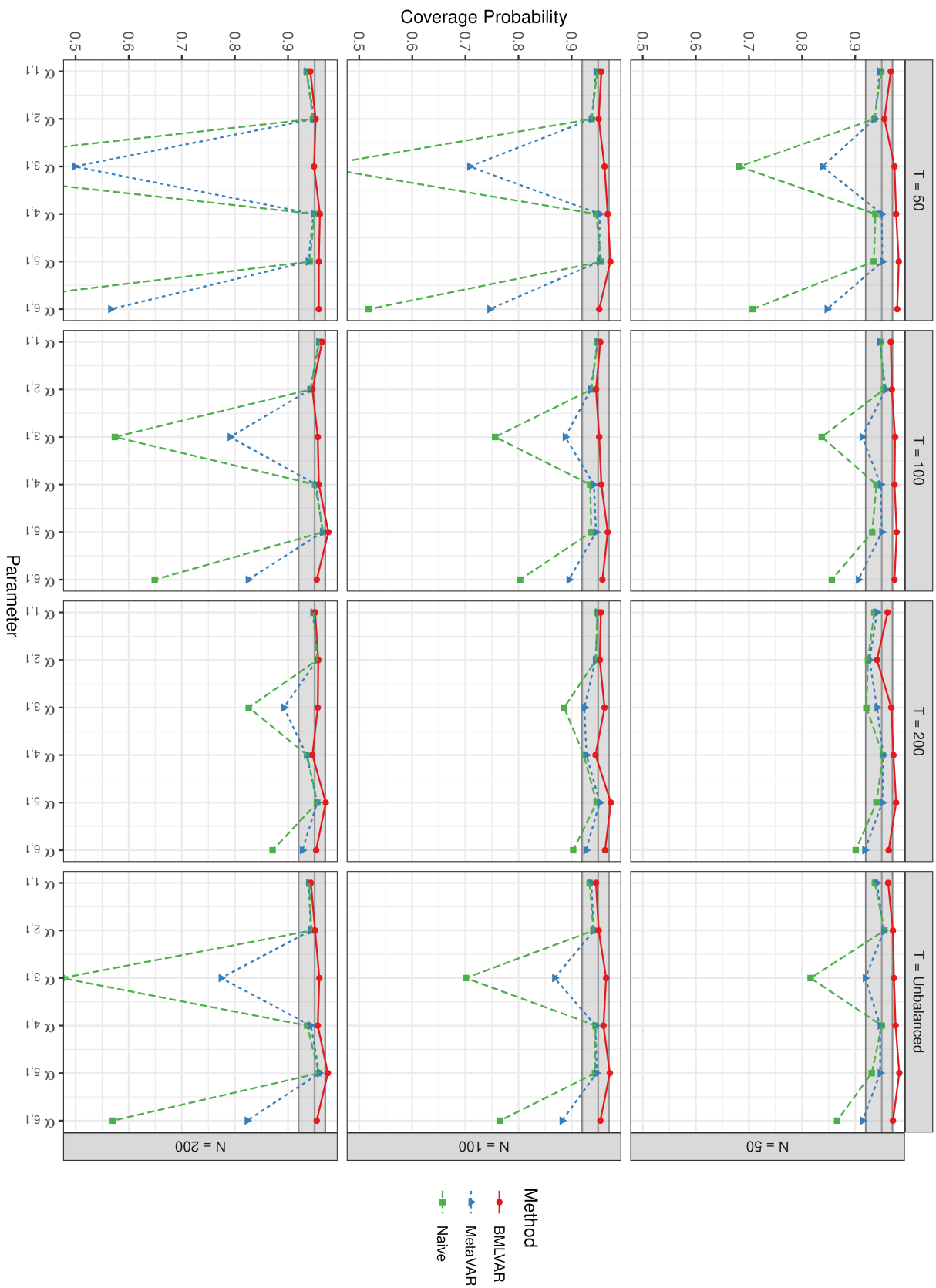


Figure 6
Coverage Probability for the Random Effects

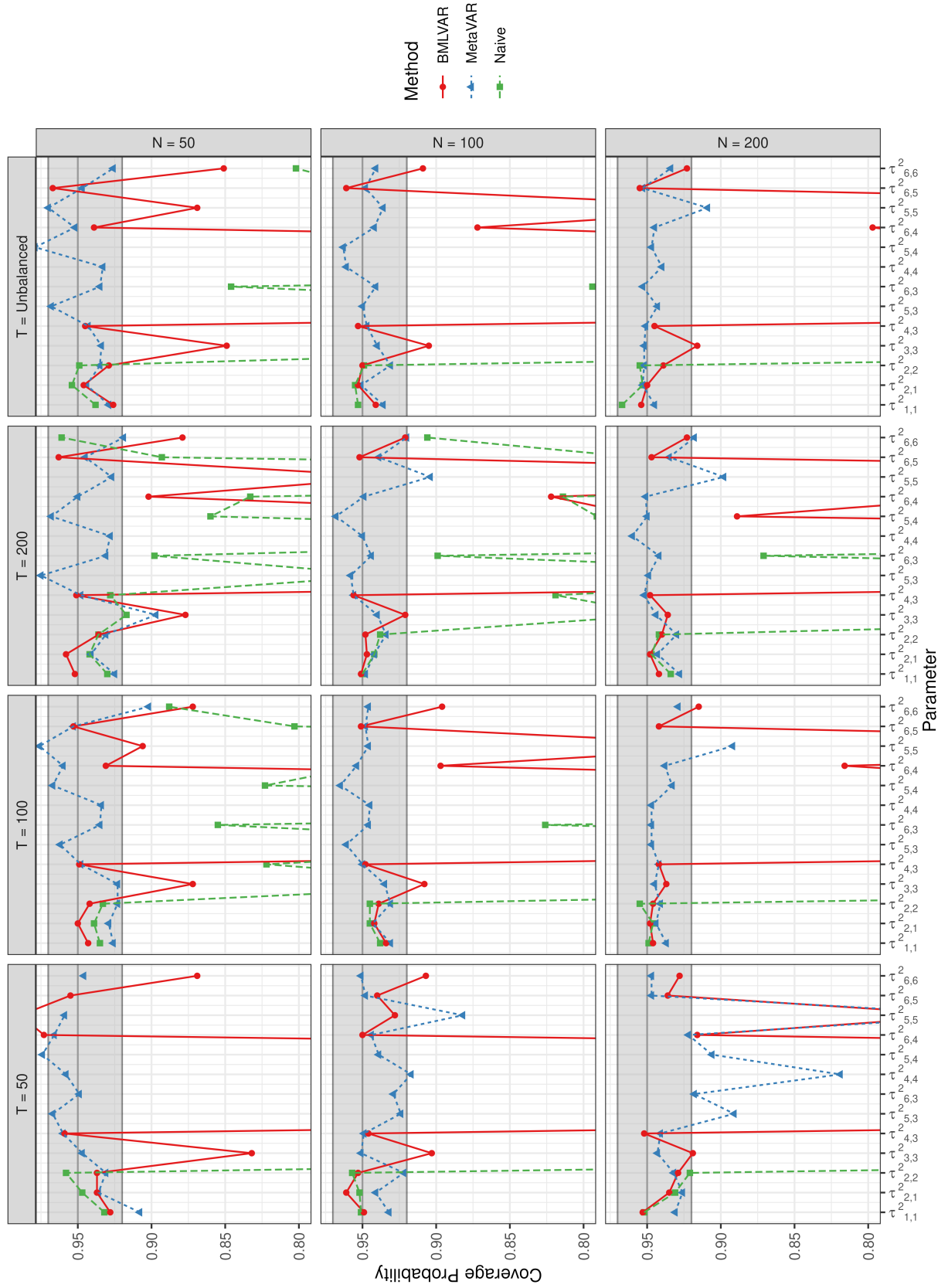


Figure 7
Statistical Power for the Fixed Effects

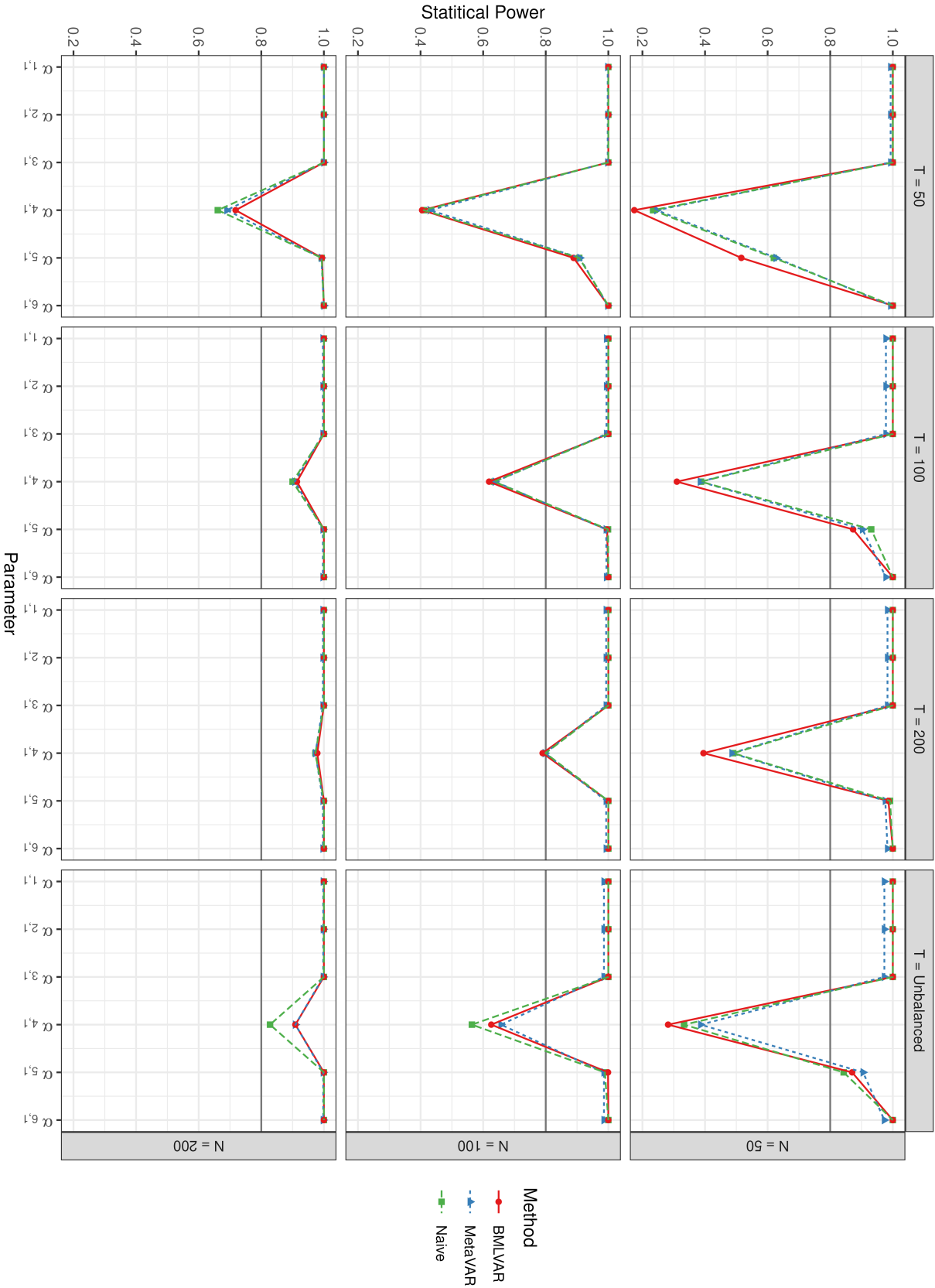


Figure 8
Statistical Power for the Random Effects

